

Jay Patil

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SUMMARY

Full Stack AI/ML Engineer with 3+ YOE and strong command of ReactJS, LLM training and fine-tuning, cloud and AI agentic frameworks. Work Authorization: F-1 student, eligible for CPT/OPT. Not subject to the \$100,000 H-1B fee.

EDUCATION

Santa Clara University

Master of Science, Computer Science and Engineering, GPA: 3.85/4.00

Sep 2024 - Jun 2026

Santa Clara, CA

MIT School of Engineering

Bachelor of Technology, Computer Science and Engineering, GPA: 3.9/4.00

May 2018 - Aug 2022

Pune, India

EXPERIENCE

Santa Clara University

AI Software Developer - Graduate Researcher

Jan 2025 - Present

Santa Clara, CA

- Fine-tuned RAG chatbot using SFT OpenAI embeddings, improving context accuracy and reducing query latency by 2s.
- Launched a Python-based semantic search portal for the SCU Fellowship Department on Firebase using vector-based retrieval and embedding indexing, replacing keyword filtering and improving search relevance by 30%
- Automated Linux/MariaDB student account provisioning with Ansible, reducing manual onboarding time by 90%

Ador Powertron

Full Stack Software Engineer

Mar 2022 - Aug 2024

Pune, India

- Co-led a team of engineers to develop concurrent Autocharge flow in JavaScript, cutting time-to-first-charge from 3 mins to 30s.
- Built Dynamic Load Balancing and Dual-VCCU switching scripts, reducing session duration by up to 50%.
- Gathered client requirements and built ReactJS dashboards with SQL database layer for real-time session monitoring and V2G-V2X battery emulation, cutting testing cycle time by 50%.
- Designed idle power consumption throttle logic, cutting idle energy consumption by ~87% (~2W per station) during peak demand.
- Deployed 20+ multithreaded microservices (OCPP 1.6) in Java using OpenAPI, enabling Smart Charging across 50+ EV stations.
- Built a full-stack IoT application in C# .NET using SMTP-NTCIP protocols to remotely control highway VMS signs, reducing reported traffic incidents from 43 to 28 annually (2024 metrics).

TECHNICAL SKILLS

- **Languages:** Python, Java, TypeScript, JavaScript, C#, C, C++, HTML5/CSS3, MongoDB, SQL
- **Full Stack:** React.js, Node.js, Streamlit, Django, Flask, Spring Boot, FastAPI, AWS, GCP, Docker, Kubernetes, Kafka, Tailwind
- **Machine Learning:** PyTorch, TensorFlow, Hugging Face, Scikit-learn, Pandas, NumPy, Weights & Biases, Deep Learning, Reinforcement Learning, LSTM, Transformers, LangChain, Ansible
- **LLMs & GenAI:** RAG, QLoRA, Fine Tuning, vLLM, Agentic Workflows, LLM Routing, Model Alignment and Evaluation

KEY PROJECTS

PulseFi LSTM

- Innovated a contactless heart rate monitor using Wi-Fi CSI signals from an ESP32, training a two-stage LSTM pipeline on 150 minutes of self-collected data across 30 sessions, achieving 5 BPM MAE within a 20s inference window at 1–2m range.

STAAR – Smart Task Aware Agentic Routing

- Engineered an agentic LLM router using a Qwen-8B SLM to classify intent and decompose multi-step prompts into parallelized sub-tasks using cost-aware routing, reducing inference latency from 8s to 3s with fewer model tokens consumed.

RobustBench-LLM

- Trained and fine-tuned a Llama-3.2-1B prompt classifier using QLoRA (4-bit NF4 quantization) and devised an adversarial robustness benchmarking suite that evaluates LLMs against GCG and AutoDAN, across 5000 prompts.

PR Manager – AI Code Review & Vulnerability Scanner

- Built a RAG-powered PR review pipeline (LangChain + Flask + Slack) that auto-generates structured code reviews and security reports on every pull request, cutting review turnaround from 45 mins to 5 mins.

Synapse AI

- Built a client-side AI meeting copilot (Grog + Whisper) with real-time transcription in a rolling context window, surfacing 3 grounded suggestions every 30s while sustaining context in multi-hour sessions.

PUBLICATIONS

- Framework for Cloud-Based Messaging System Using MQTT - IEEE, 2020.
Designed a scalable AWS IoT pub-sub architecture in Python enabling real-time bidirectional device telemetry.
- Development of a Comparison Based Medicine Purchasing System.
Implemented a Selenium based price comparison engine using Python for pharmacy platforms, ranking drug affordability.